

where the mean and unbiased variance are estimated as:

$$\mathbb{R} \ni \hat{\mu}_k = \frac{1}{n_k} \sum_{i=1}^n x_{i,j} \mathbb{I}(\mathcal{C}(x_i) = k), \quad (64)$$

$$\mathbb{R} \ni \hat{\sigma}_k^2 = \frac{1}{n_k - 1} \sum_{i=1}^n (x_{i,j} - \hat{\mu}_k)^2 \mathbb{I}(\mathcal{C}(x_i) = k), \quad (65)$$

where $x_{i,j}$ denotes the j -th feature of the i -th training instance. The prior can again be estimated using Eq. (32).

According to Eqs. (62) and (63), Gaussian naive Bayes is equivalent to QDA where the covariance matrices are *diagonal*, i.e., the off-diagonal of the covariance matrices are ignored. Therefore, we can say that QDA is more powerful than Gaussian naive Bayes because Gaussian naive Bayes is a simplified version of QDA. Moreover, it is obvious that Gaussian naive Bayes and QDA are equivalent for one dimensional data. Comparing to LDA, the Gaussian naive Bayes is equivalent to LDA if the covariance matrices are diagonal and they are all equal, i.e., $\sigma_1^2 = \dots = \sigma_{|C|}^2$; therefore, LDA and Gaussian naive Bayes have their own assumptions, one on the off-diagonal of covariance matrices and the other one on equality of the covariance matrices. As Gaussian naive Bayes has some level of optimality (Zhang, 2004), it becomes clear why LDA and QDA are such effective classifiers.

11. Relation to Likelihood Ratio Test

Consider two hypotheses for estimating some parameter, a null hypothesis H_0 and an alternative hypothesis H_A . The probability $\mathbb{P}(\text{reject } H_0 | H_0)$ is called type 1 error, false positive error, or false alarm error. The probability $\mathbb{P}(\text{accept } H_0 | H_A)$ is called type 2 error or false negative error. The $\mathbb{P}(\text{reject } H_0 | H_0)$ is also called *significance level*, while $1 - \mathbb{P}(\text{accept } H_0 | H_A) = \mathbb{P}(\text{reject } H_0 | H_A)$ is called *power*.

If $L(\theta_A)$ and $L(\theta_0)$ are the likelihoods (probabilities) for the alternative and null hypotheses, the likelihood ratio is:

$$\Lambda = \frac{L(\theta_A)}{L(\theta_0)} = \frac{f(\mathbf{x}; \theta_A)}{f(\mathbf{x}; \theta_0)}. \quad (66)$$

The Likelihood Ratio Test (LRT) (Casella & Berger, 2002) rejects the H_0 in favor of H_A if the likelihood ratio is greater than a threshold, i.e., $\Lambda \geq t$. The LRT is a very effective statistical test because according to the Neyman-Pearson lemma (Neyman & Pearson, 1933), it has the largest power among all statistical tests with the same significance level.

If the sample size is large, $n \rightarrow \infty$, and the θ_A is estimated using MLE, the logarithm of the likelihood ratio asymptotically has the distribution of χ^2 under the null hypothesis (White, 1984; Casella & Berger, 2002):

$$2 \ln(\Lambda) \overset{H_0}{\sim} \chi_{(df)}^2, \quad (67)$$

where the degree of freedom of χ^2 distribution is $df := \dim(H_A) - \dim(H_0)$ and $\dim(\cdot)$ is the number of unspecified parameters in the hypothesis.

There is a connection between LDA or QDA and the LRT (Lachenbruch & Goldstein, 1979). Recall Eq. (12) or (15) which can be restated as:

$$\frac{f_2(\mathbf{x}) \pi_2}{f_1(\mathbf{x}) \pi_1} = 1, \quad (68)$$

which is for the decision boundary. The Eq. (22) dealt with the difference of $f_2(x) \pi_2$ and $f_1(x) \pi_1$; however, here we are dealing with their ratio. Recall Fig. 1 where if we move x^* to the right and left, the ratio $f_2(x^*) \pi_2 / f_1(x^*) \pi_1$ decreases and increases, respectively, because the probabilities of the first and second class happening change. In other words, moving the x^* changes the significance level and power. Therefore, Eq. (68) can be used to have a statistical test where the posteriors are used in the ratio, as we also used posteriors in LDA and QDA. The null/alternative hypothesis can be considered to be the mean and covariance of the first/second class. In other words, the two hypotheses say that the point belongs to a specific class. Hence, if the ratio is larger than a value t , the instance $\mathbf{x}$ is estimated to belong to the second class; otherwise, the first class is chosen. According to Eq. (16), the Eq. (68) becomes:

$$\frac{(|\Sigma_2|)^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_2)^\top \Sigma_2^{-1}(\mathbf{x} - \boldsymbol{\mu}_2)\right) \pi_2}{(|\Sigma_1|)^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_1)^\top \Sigma_1^{-1}(\mathbf{x} - \boldsymbol{\mu}_1)\right) \pi_1} \geq t, \quad (69)$$

for QDA. In LDA, the covariance matrices are equal, so:

$$\frac{\exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_2)^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}_2)\right) \pi_2}{\exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_1)^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}_1)\right) \pi_1} \geq t. \quad (70)$$

As can be seen, changing the priors change impacts the ratio as expected. Moreover, the value of t can be chosen according to the desired significance level in the χ^2 distribution using the χ^2 table. The Eqs. (69) and (70) show the relation of LDA and QDA with LRT. As the LRT has the largest power (Neyman & Pearson, 1933), the effectiveness of LDA and QDA in classification is explained from a hypothesis testing point of view.

12. Simulations

In this section, we report some simulations which make the concepts of tutorial clearer by illustration.

12.1. Experiments with Equal Class Sample Sizes

We created a synthetic dataset of three classes each of which is a two dimensional Gaussian distribution. The means and covariance matrices of the three Gaussians from

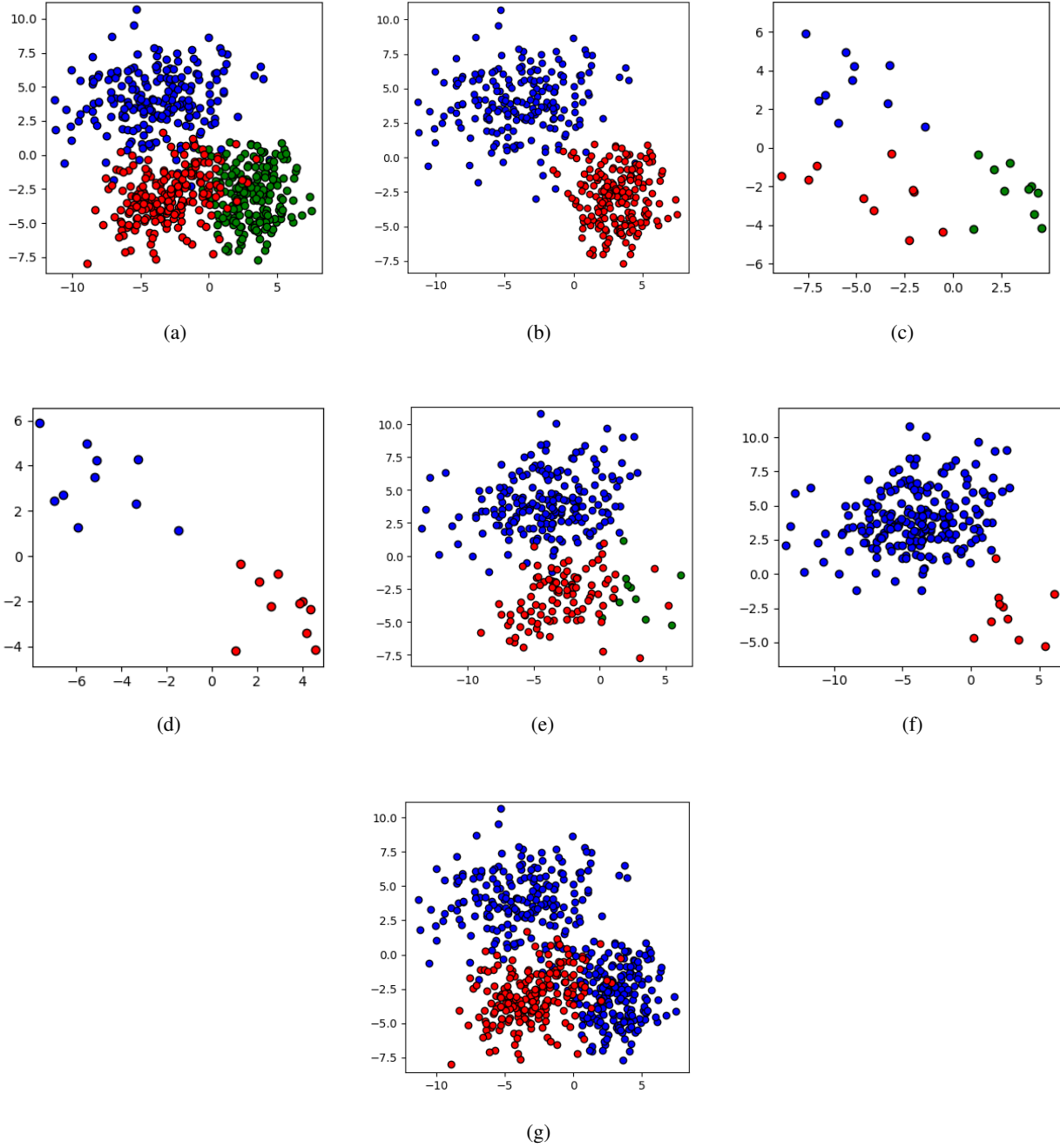


Figure 3. The synthetic dataset: (a) three classes each with size 200, (b) two classes each with size 200, (c) three classes each with size 10, (d) two classes each with size 10, (e) three classes with sizes 200, 100, and 10, (f) two classes with sizes 200 and 10, and (g) two classes with sizes 400 and 200 where the larger class has two modes.

which the class samples were randomly drawn are:

$$\begin{aligned} \mu_1 &= [-4, 4]^\top, \quad \mu_2 = [3, -3]^\top, \quad \mu_3 = [-3, 3]^\top, \\ \Sigma_1 &= \begin{bmatrix} 10 & 1 \\ 1 & 5 \end{bmatrix}, \quad \Sigma_2 = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}, \quad \Sigma_3 = \begin{bmatrix} 6 & 1.5 \\ 1.5 & 4 \end{bmatrix}. \end{aligned}$$

The three classes are shown in Fig. 3-a where each has sample size 200. Experiments were performed on the three classes. We also performed experiments on two of the three classes to test a binary classification. The two classes are shown in Fig. 3-b. The LDA, QDA, naive Bayes, and

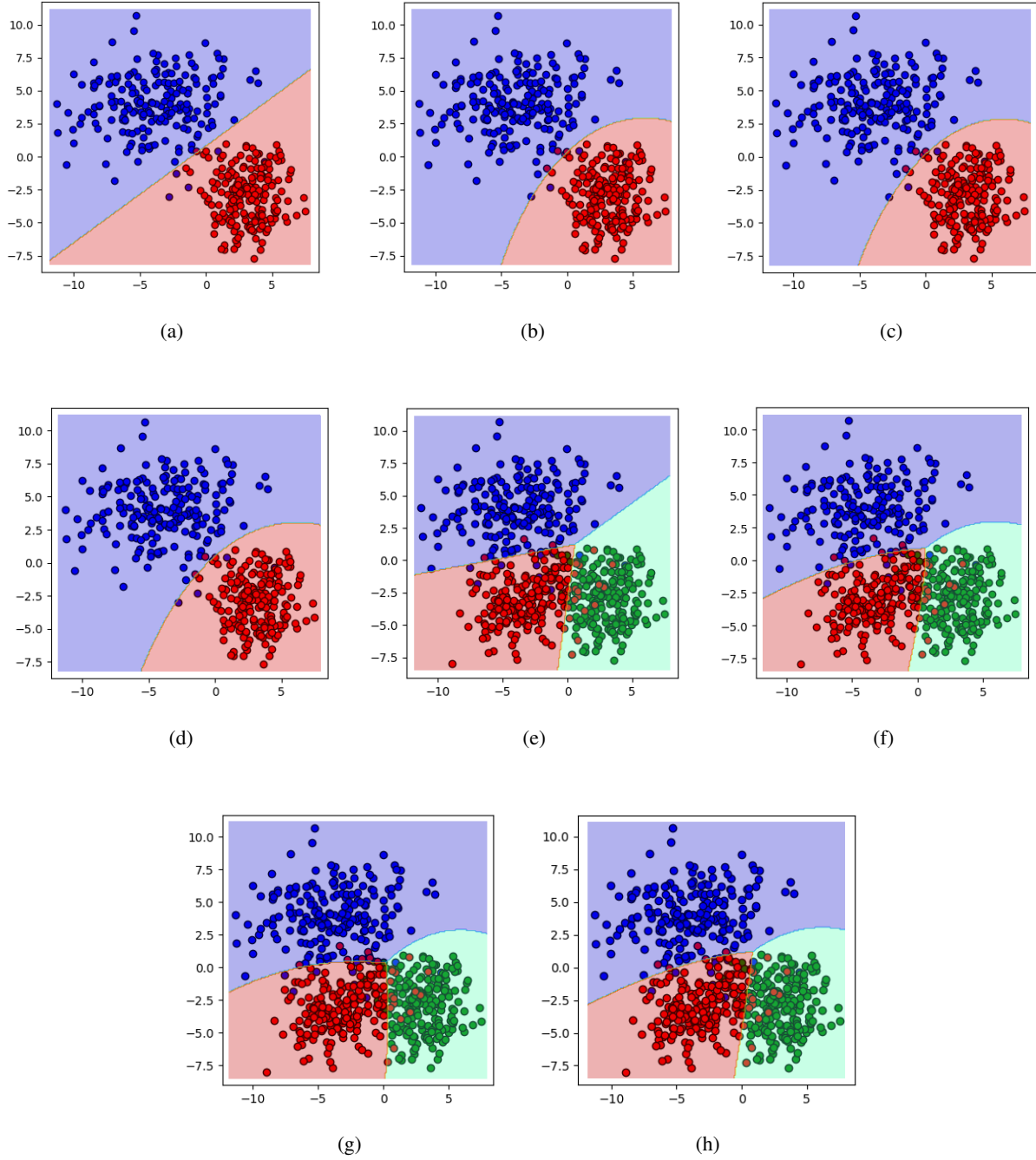


Figure 4. Experiments with equal class sample sizes: (a) LDA for two classes, (b) QDA for two classes, (c) Gaussian naive Bayes for two classes, (d) Bayes for two classes, (e) LDA for three classes, (f) QDA for three classes, (g) Gaussian naive Bayes for three classes, and (h) Bayes for three classes.

Bayes classifications of the two and three classes are shown in Fig. 4. For both binary and ternary classification with LDA and QDA, we used Eqs. (31) and (28), respectively, with Eq. (29). We also estimated the mean and covariance using Eqs. (33), (35), and (36). For Gaussian naive Bayes,

we used Eqs. (62) and (63) and estimated the parameters using Eqs. (64) and (65). For Bayes classifier, we used Eq. (58) with Eq. (63) but we do not estimate the mean and variance; except, in order to use the exact likelihoods in Eq. (58), we use the exact mean and covariance matrices

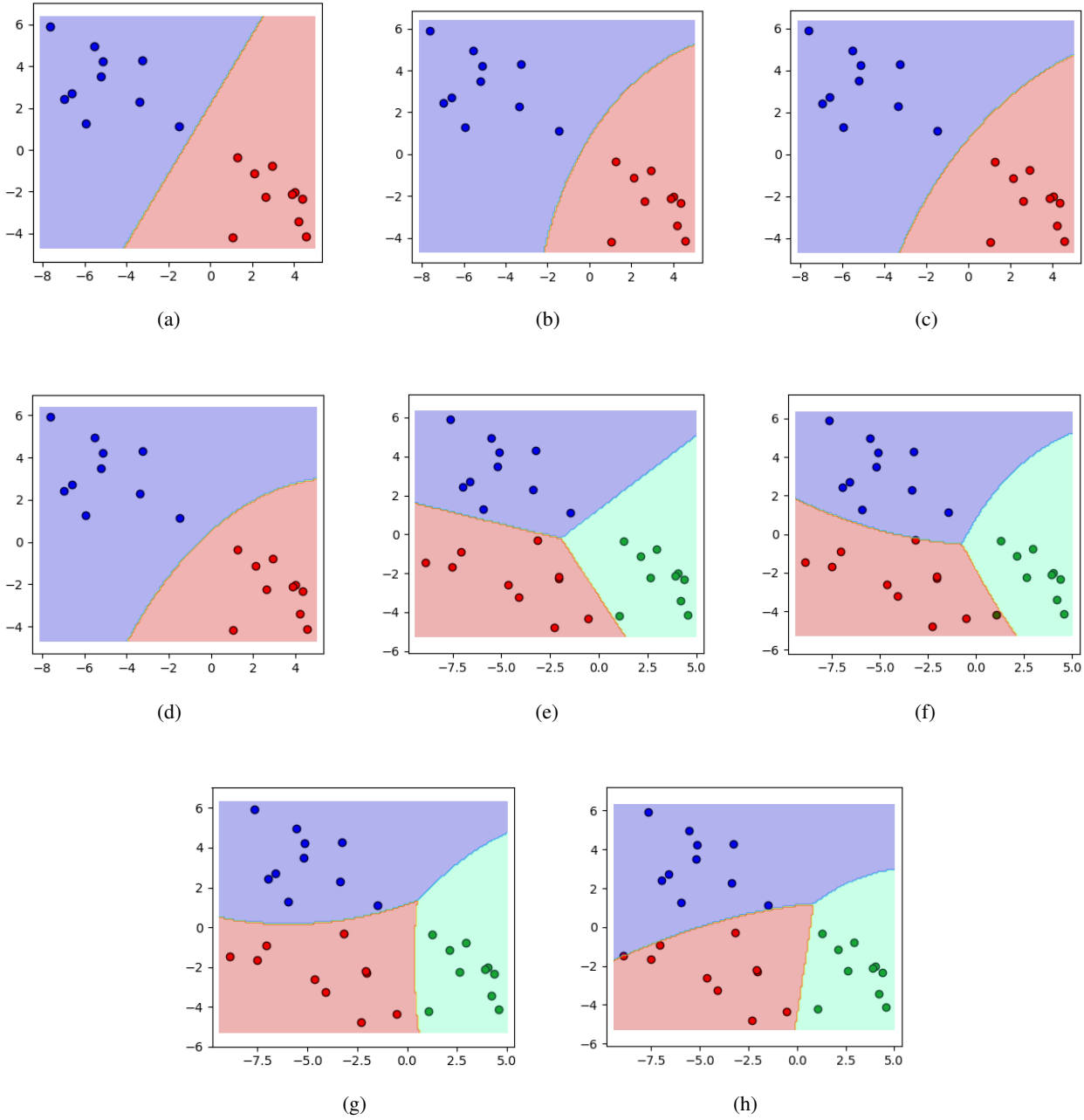


Figure 5. Experiments with small class sample sizes: (a) LDA for two classes, (b) QDA for two classes, (c) Gaussian naive Bayes for two classes, (d) Bayes for two classes, (e) LDA for three classes, (f) QDA for three classes, (g) Gaussian naive Bayes for three classes, and (h) Bayes for three classes.

of the distributions which we sampled from. We, however, estimated the priors. The priors were estimated using Eq. (32) for all the classifiers.

As can be seen in Fig. 4, the space is partitioned into two/three parts and this validates the assertion that LDA and QDA can be considered as metric learning methods as discussed in Section 7. As expected, the boundaries of

LDA and QDA are linear and curvy (quadratic), respectively. The results of QDA, Gaussian naive Bayes, and Bayes are very similar although they have slight differences. This is because the classes are already Gaussian so if the estimates of means and covariance matrices are accurate enough, QDA and Bayes are equivalent. The classes are Gaussian and the off-diagonal elements of covariance

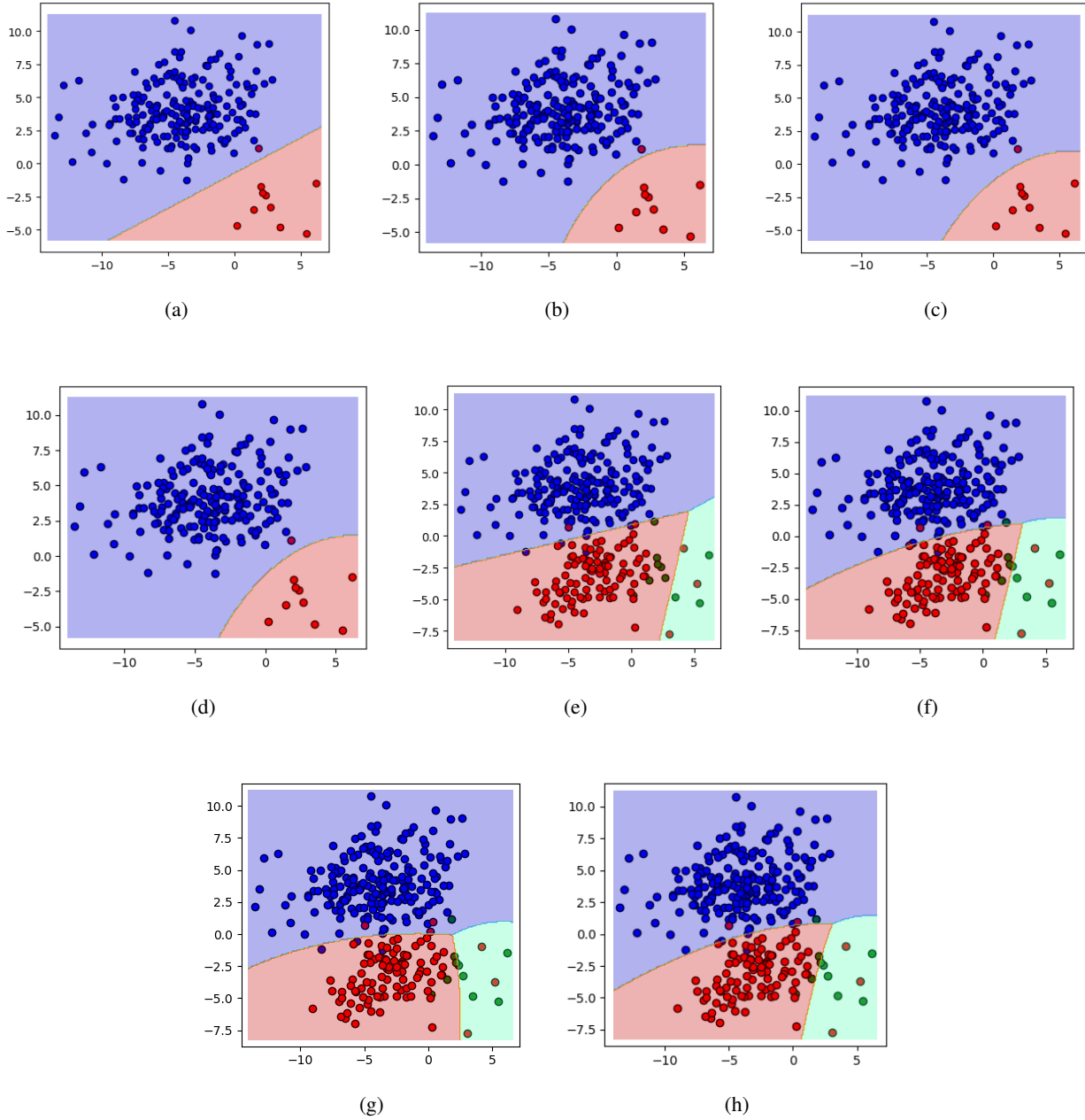


Figure 6. Experiments with different class sample sizes: (a) LDA for two classes, (b) QDA for two classes, (c) Gaussian naive Bayes for two classes, (d) Bayes for two classes, (e) LDA for three classes, (f) QDA for three classes, (g) Gaussian naive Bayes for three classes, and (h) Bayes for three classes.

matrices are also small compared to the diagonal; therefore, naive Bayes is also behaving similarly.

12.2. Experiments with Small Class Sample Sizes

According to Monte-Carlo approximation (Robert & Casella, 2013), the estimates in Eqs. (33), (35), (64) and (65) are more accurate if the sample size goes to infinity,

i.e., $n \rightarrow \infty$. Therefore, if the sample size is small, we expect mode difference between QDA and Bayes classifiers. We made a synthetic dataset with three or two classes with the same mentioned means and covariance matrices. The sample size of every class was 10. Figures 3-c and 3-d show these datasets. The results of LDA, QDA, Gaussian naive Bayes, and Bayes classifiers for this dataset are

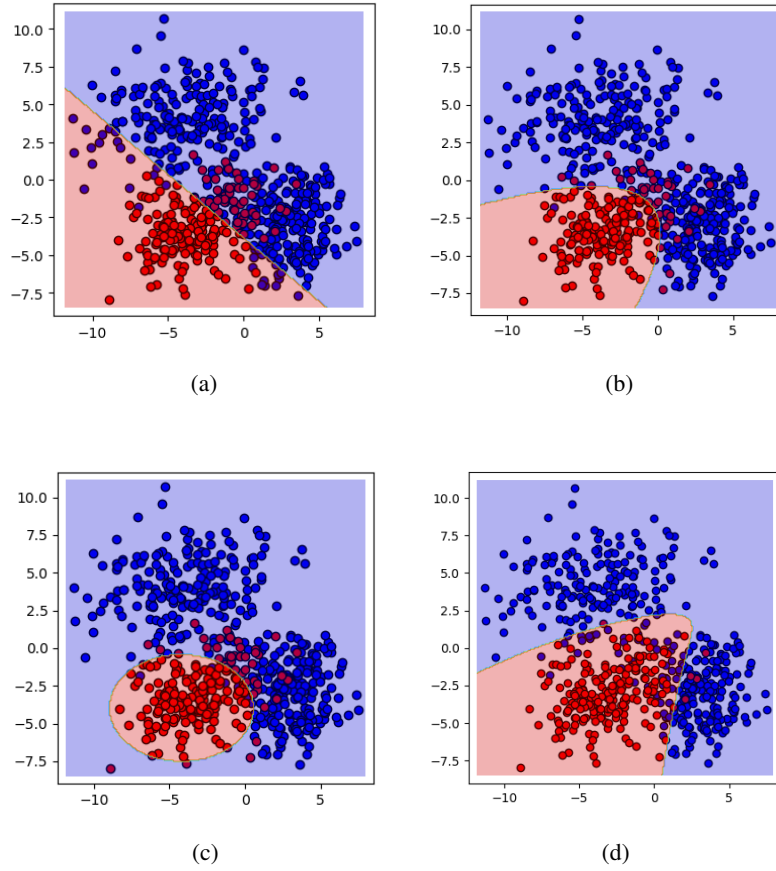


Figure 7. Experiments with multi-modal data: (a) LDA, (b) QDA, (c) Gaussian naive Bayes, and (d) Bayes.

shown in Fig. 5. As can be seen, now, the results of QDA, Gaussian naive Bayes, and Bayes are different for the reason explained.

12.3. Experiments with Different Class Sample Sizes

According to Eq. (32) used in Eqs. (28), (31), (58), and (62), the prior of a class changes by the sample size of the class. In order to see the effect of sample size, we made a synthetic dataset with different class sizes, i.e., 200, 100, and 10, shown in Figs. 3-e, 3-f. We used the same mentioned means and covariance matrices. The results are shown in Fig. 6. As can be seen, the class with small sample size has covered a small portion of space in discrimination which is expected because its prior is small according to Eq. (32); therefore, its posterior is small. On the other hand, the class with large sample size has covered a larger portion because of a larger prior.

12.4. Experiments with Multi-Modal Data

As mentioned in Section 8, LDA and QDA assume uni-modal Gaussian distribution for every class and thus FDA

or LDA faces problem for multi-modal data (Sugiyama, 2007). For testing this, we made a synthetic dataset with two classes, one with sample size 400 having two modes of Gaussians and the other with sample size 200 having one mode. We again used the same mentioned means and covariance matrices. The dataset is shown in Fig. 3-g.

The results of the LDA, QDA, Gaussian naive Bayes, and Bayes classifiers for this dataset are shown in Fig. 7. The mean and covariance matrix of the larger class, although it has two modes, were estimated using Eqs. (33), (35), (64) and (65) in LDA, QDA, and Gaussian naive Bayes. However, for the likelihood used in Bayes classifier, i.e., in Eq. (58), we need to know the exact multi-modal distribution. Therefore, we fit a mixture of two Gaussians (Ghosh et al., 2019a) to the data of the larger class:

$$\mathbb{P}(X = \mathbf{x} \mid \mathbf{x} \in \mathcal{C}_k) = \sum_{k=1}^2 w_k f(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad (71)$$

where $f(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is Eq. (16) and we the fitted parame-

ters were:

$$\begin{aligned}\mu_1 &= [-3.88, 4]^\top, \quad \mu_2 = [3.04, -2.92]^\top, \\ \Sigma_1 &= \begin{bmatrix} 9.27 & 0.79 \\ 0.79 & 4.82 \end{bmatrix}, \quad \Sigma_2 = \begin{bmatrix} 2.87 & 0.03 \\ 0.03 & 3.78 \end{bmatrix}, \\ w_1 &= 0.49, \quad w_2 = 0.502.\end{aligned}$$

As Fig. 7 shows, LDA has not performed well enough as expected. The performance of QDA is more acceptable than LDA but still not good enough because QDA also assumes a uni-modal Gaussian for every class. The result of Gaussian naive Bayes is very different from the Bayes here because the Gaussian naive Bayes assumes uni-modal Gaussian with diagonal covariance for every class. Finally, the Bayes has the best result as it takes into account the multi-modality of the data and it is optimum (Mitchell, 1997).

13. Conclusion and Future Work

This paper was a tutorial paper for LDA and QDA as two fundamental classification methods. We explained the relations of these two methods with some other methods in machine learning, manifold (subspace) learning, metric learning, statistics, and statistical testing. Some simulations were also provided for better clarification.

This paper focused on LDA and QDA which are discriminators with one and two polynomial degrees of freedom, respectively. As the future work, we will work on a tutorial paper for non-linear discriminant analysis using kernels (Baudat & Anouar, 2000; Li et al., 2003; Lu et al., 2003), which is called *kernel discriminant analysis*, to have discriminators with more than two degrees of freedom.

Acknowledgment

The authors hugely thank Prof. Ali Ghodsi (see his great online related courses (Ghodsi, 2015; 2017)), Prof. Mu Zhu, Prof. Hoda Mohammadzade, and other professors whose courses have partly covered the materials mentioned in this tutorial paper.

References

- Baudat, Gaston and Anouar, Fatiha. Generalized discriminant analysis using a kernel approach. *Neural computation*, 12(10):2385–2404, 2000.
- Belhumeur, Peter N, Hespanha, João P, and Kriegman, David J. Eigenfaces vs. Fisherfaces: Recognition using class specific linear projection. *IEEE Transactions on Pattern Analysis & Machine Intelligence*, (7):711–720, 1997.
- Boyd, Stephen and Vandenberghe, Lieven. *Convex optimization*. Cambridge university press, 2004.
- Casella, George and Berger, Roger L. *Statistical inference*, volume 2. Duxbury Pacific Grove, CA, 2002.
- Cox, Trevor F and Cox, Michael AA. *Multidimensional scaling*. Chapman and hall/CRC, 2000.
- Croot, Ernie. The Rayleigh principle for finding eigenvalues. Technical report, Georgia Institute of Technology, School of Mathematics, 2005. Online: http://people.math.gatech.edu/~ecroot/notes_linear.pdf, Accessed: March 2019.
- De Maesschalck, Roy, Jouan-Rimbaud, Delphine, and Massart, Désiré L. The Mahalanobis distance. *Chemo-metrics and intelligent laboratory systems*, 50(1):1–18, 2000.
- Etemad, Kamran and Chellappa, Rama. Discriminant analysis for recognition of human face images. *Journal of the Optical Society of America A*, 14(8):1724–1733, 1997.
- Fisher, Ronald A. The use of multiple measurements in taxonomic problems. *Annals of eugenics*, 7(2):179–188, 1936.
- Friedman, Jerome, Hastie, Trevor, and Tibshirani, Robert. *The elements of statistical learning*, volume 2. Springer series in statistics, New York, NY, USA, 2009.
- Ghodsi, Ali. Classification course, department of statistics and actuarial science, university of Waterloo. Online Youtube Videos, 2015. Accessed: January 2019.
- Ghodsi, Ali. Data visualization course, department of statistics and actuarial science, university of Waterloo. Online Youtube Videos, 2017. Accessed: January 2019.
- Ghojogh, Benyamin, Mohammadzade, Hoda, and Mokari, Mozghan. Fisherposes for human action recognition using Kinect sensor data. *IEEE Sensors Journal*, 18(4):1612–1627, 2017.
- Ghojogh, Benyamin, Ghojogh, Aydin, Crowley, Mark, and Karray, Fakhri. Fitting a mixture distribution to data: Tutorial. *arXiv preprint arXiv:1901.06708*, 2019a.
- Ghojogh, Benyamin, Karray, Fakhri, and Crowley, Mark. Eigenvalue and generalized eigenvalue problems: Tutorial. *arXiv preprint arXiv:1903.11240*, 2019b.
- Ham, Ji Hun, Lee, Daniel D, Mika, Sebastian, and Schölkopf, Bernhard. A kernel view of the dimensionality reduction of manifolds. In *International Conference on Machine Learning*, 2004.
- Hazewinkel, Michiel. Central limit theorem. *Encyclopedia of Mathematics*, Springer, 2001.
- Jolliffe, Ian. *Principal component analysis*. Springer, 2011.